

A Quantitative Theory of the Rise of Intangibles, Multinational Production, and Profit Shifting*

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March 20, 2025

Work in Progress

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Abstract

Over the past three decades, the global economy has experienced three key trends: the rise of intangible capital, the expansion of multinational enterprises (MNEs), and the increasing prevalence of international profit shifting. In this paper, we demonstrate that these trends are both empirically and structurally interconnected, reinforcing one another over time. Using cross-sectional evidence and event studies, we show that intangible intensity covaries positively with the scale of MNE activities and firm-level profit-shifting indicators. To rationalize these relationships, we develop a general equilibrium model that highlights scalable complementarity: reductions in intangible costs and trade barriers together amplify intangible investment and multinational expansion, with these effects further magnified by the nonrival nature of intangible capital. Profit shifting, in turn, rises superlinearly as tax differentials, enforcement costs, globalization, and intangible scalability shape firm behavior. Through counterfactual analysis, we quantify the contributions of technological change, globalization, and tax policy, demonstrating that their interdependence contributes significantly to the rise of intangible-intensive MNEs engaged in profit shifting.

Keywords: Multinational production; intangible capital; profit shifting; corporate taxation.

JEL Codes: E6, F23, H25, H26, H3

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1 Introduction

Over the past four decades, two key trends have dramatically reshaped the global economy: the rise of intangible capital and the expansion of multinational production. We argue that these trends are connected due to intangible capital’s nonrivalry: Unlike tangible capital, intellectual property, software, other forms of intangible capital can be deployed simultaneously in multiple productive establishments. A multinational firm has greater incentive to invest in intangible capital to exploit this nonrivalry than a non-MNE, so the former will have a higher intangible intensity. At the same time, a firm with a higher intangible intensity has a greater incentive to establish foreign affiliates than a firm with a lower intangible intensity, so the former is more likely to engage in multinational production. Consequently, reductions in the barriers to multinational production will increase intangible intensity, and intangible-biased changes in production technology or reductions in the cost of producing intangible capital will increase multinational production. We use a quantitative model to measure the contributions of these exogenous driving forces to these two trends.

Additionally, we explore the role of a third trend: the rise of international profit shifting by multinationals. Empirical evidence indicates that intangible capital plays a critical role in profit shifting, as it is easy to legally transfer it across borders and manipulate the transfer pricing of the income that accrues to it. As shown by [Dyrda, Hong, and Steinberg \(2024a\)](#), profit shifting raises MNEs’ incentives to invest in intangible capital. At the same time, however, increased intangible intensity creates more opportunity for MNEs to shift profits. We extend our model to incorporate profit shifting and decompose the observed endogenous changes over time in intangible intensity, multinational production, and profit shifting into exogenous changes in production technologies, barriers to multinational production, and technological costs of profit shifting.

Our first contribution is to document these three interconnected trends using linked firm-level data from Compustat and U.S. Securities and Exchange Commission (SEC) disclosures. First, intangible intensity—measured as the ratio of intangible assets (including capitalized R&D) to total assets—more than doubled among U.S. public firms between 1995 and 2023, rising from 22% to 42%. This growth occurred uniformly across industries, driven by within-industry increases rather than sectoral reallocation. Meanwhile, multinational activity surged, with the share of firms operating foreign subsidiaries increasing from 16% to 38% and the average foreign sales share of MNEs rising from 10% to 27%. Profit shifting, while proxied by tax-haven subsidiary ownership, grew in parallel, peaking at 74% of MNEs in 2017 before declining post-TCJA. Cross-sectional regressions reveal that MNEs exhibit 11.1 percentage points higher intangible intensity than domestic firms, a gap that widens post-2005. Firms operating in more countries or engaging in profit shift-

ing show even greater intangible intensity. While these patterns hold after controlling for firm size, year fixed effects, and industry composition, the relationships are inherently correlational, as unobserved heterogeneity (e.g., productivity, strategic priorities) and reverse causality (e.g., high intangible intensity incentivizing multinational expansion) complicate causal interpretation.

To explore dynamics over time, we employ an event-study framework analyzing firms before and after multinational entry or initial profit-shifting activity. Firms that become MNEs experience a 4 percentage point rise in intangible intensity post-entry, while those initiating profit shifting see a smaller, transient 3 percentage point increase. However, these associations should not be interpreted causally, as firms’ decisions to expand or shift profits are endogenous—driven by unobservables like growth opportunities or tax optimization strategies. For instance, firms anticipating higher returns to intangibles may self-select into multinational production, creating a spurious link between MNE status and intangible intensity. These limitations motivate our structural model, which disentangles bidirectional mechanisms (e.g., nonrivalry-driven FDI vs. FDI-induced intangible accumulation) and accounts for endogenous firm choices.

Motivated by these empirical patterns—and their inherent limitations in identifying causal mechanisms—we develop a macroeconomic framework to structurally link the rise of intangibles, multinational expansion, and profit shifting. This framework, our second contribution, addresses the endogeneity and reverse causality challenges inherent in the data by explicitly modeling firms’ endogenous decisions to invest in intangibles, expand globally, and shift profits. The model is a multi-region general equilibrium with households, final goods producers, intermediate goods producers, and governments. Central to the analysis are intermediate goods producers, which are headquartered in one region and choose where to export or establish foreign subsidiaries (FDI). Critically, production combines locally sourced tangible capital and labor with *nonrival intangible capital* developed at the headquarters.

We formalize nonrivalry as a parameter that governs a firm’s ability to deploy intangible capital across subsidiaries simultaneously. For example, a software firm’s code can be reused globally (high nonrivalry), whereas a manufacturer’s process patents are more location-specific (low nonrivalry). Subsidiaries use intangibles with efficiency proportional to this nonrivalry parameter, creating endogenous incentives for multinational expansion. Building on our prior work ([Dyrda et al., 2024a](#)), we integrate profit shifting through transfer pricing: subsidiaries pay licensing fees for intangible use, but MNEs can reallocate ownership of these assets—at a cost—to affiliates in low-tax jurisdictions. This allows profits to be shifted from high-tax parents to tax havens, raising after-tax returns. Embedding these mechanisms into a unified framework allows us study jointly

the driving forces of the observed trends both theoretically and quantitatively.

Our model identifies two fundamental mechanisms that structurally explain and link the empirical trends. First, reductions in the costs of producing intangible capital (e.g., cheaper R&D or digital tools) and declines in barriers to multinational expansion (e.g., trade and FDI liberalization) act as scalable complements. Each factor independently raises intangible intensity and firms' propensity to operate abroad, but their joint effect is superlinear: the combined impact exceeds the sum of their isolated effects. This occurs because lower intangible costs incentivize firms to invest in scalable assets, while reduced trade and FDI barriers amplify the returns to deploying those assets globally. Critically, the strength of these responses depends on the degree of intangible nonrivalry—the ability to reuse intangibles across subsidiaries. Firms with highly nonrival intangibles (e.g., tech firms) exhibit the strongest reactions, as their assets gain value when scaled across borders. This mechanism rationalizes why sectors with scalable intangibles drove the rise of globalized, intangible-intensive MNEs.

Second, profit shifting emerges endogenously from the interplay of tax differentials, technological shifting costs, and intangible scalability. A widening gap between home-country and tax-haven rates, alongside falling costs of relocating intellectual property (e.g., weaker enforcement), incentivizes profit reallocation. These forces also interact superlinearly with intangible costs and FDI barriers: for example, cheaper intangibles raise the value of shifted profits, while globalization expands opportunities to exploit tax gaps. Again, nonrivalry amplifies these effects—firms with scalable intangibles shift profits more aggressively because their assets generate cross-border income at minimal marginal cost. Together, the model demonstrates how technological advances (lower intangible costs), policy shifts (trade liberalization, tax reforms), and firm-level nonrivalry jointly drive the observed rise in intangible intensity, multinational expansion, and profit shifting. These trends are not isolated phenomena but mutually reinforcing outcomes of a unified structural process, resolving the endogeneity challenges inherent in purely empirical analyses.

Our quantitative analysis (still in progress), which we view as third contribution of this paper, proceeds in three integrated stages, designed to discipline the model inputs, validate the model, and isolate the roles of technological change, globalization, and tax incentives on empirical trends. In the first stage we calibrate time-varying primitives between 1995-2023 so that they can serve as inputs to the analysis. Intangible production costs are mapped to sector-country trends in R&D expenditures, digitalization indices, and software prices. Trade/FDI barriers are calibrated using bilateral FDI restrictiveness indices, tariff schedules, and services trade barriers. Tax differentials and profit-shifting costs reflect statutory tax rates and country-specific enforcement measures.

Firm-level heterogeneity in intangible nonrivalry is pinned down by cross-sectional Compustat patterns—e.g., tech firms exhibit stronger links between intangible intensity and multinational expansion. The model’s multi-country structure (20+ regions, including all major Compustat subsidiary locations and tax havens) ensures granular mapping to firm-level disclosures. Following this calibration strategy we force the model to match the observed trends *by design*.

In the second stage, we validate the model by testing its ability to replicate both the cross-sectional correlations and dynamic patterns documented in the data. We test whether our framework successfully matches the strong positive relationships between intangible intensity, multinational status, and profit shifting—including the post-2005 acceleration in these links—as well as the temporal alignment between multinational entry, profit-shifting initiation, and intangible growth observed in event studies. This is a critically important test, which illustrates whether the model replicates, through structural relationships, the magnitude of interdependence between three trends that we found in the data. We also use the Tax Cuts and Jobs Act (TCJA) to further validate our model. The TCJA introduced tax provisions for MNEs—including Global Intangible Low-Taxed Income (GILTI), Foreign-Derived Intangible Income (FDII), and the Base Erosion and Anti-Abuse Tax (BEAT)—designed to reduce profit-shifting incentives. Our analysis of aggregate trends shows that after the TCJA was implemented, the share of profit-shifting firms declined significantly, while the growth of intangible intensity and multinational expansion slowed. Using firm-level data and a difference-in-differences design, we provide causal evidence that MNEs affected by the TCJA not only reduced profit shifting but also curtailed intangible investment and multinational production. These findings complement prior research on the TCJA’s impact on tangible investment ([Chodorow-Reich, Smith, Zidar, and Zwick, 2023](#)). Building on our earlier work in [Dyrda, Hong, and Steinberg \(2024b\)](#), we incorporate the TCJA framework into our model and assess its effects on firms’ decisions regarding intangible investment and global production.

In the third stage, we conduct counterfactual simulations to decompose the contributions of technological change, globalization, and tax incentives—both individually and in tandem—to the observed trends. We first simulate scenarios where each driver is held fixed at 1995 levels (e.g., no decline in intangible costs, static trade barriers, or unchanged tax differentials). To uncover interdependence, we then examine pairwise combinations: for instance, allowing intangible costs and globalization to evolve jointly while fixing tax incentives, or pairing tax shifts with falling intangible costs while halting trade liberalization. These exercises reveal superlinear interactions—where the combined impact of two drivers exceeds the sum of their isolated effects—demonstrating how falling intangible costs amplify globalization’s role in multinational expansion, or how tax compe-

tition magnifies the returns to scalable intangibles. Finally, a "stagnant world" counterfactual (all drivers fixed but factor neutral TFP growth operates) benchmarks the cumulative effect of post-1995 changes. By comparing these scenarios, we quantify how much of the rise in intangible intensity, multinational activity, and profit shifting stems from each driver's direct impact, versus synergies between them. The results of this quantification are pending.

The rest of the paper is organized as follows. Section 2 describes empirical evidence. We present our quantitative theory in Section 3. Section 4 presents theoretical implications of our model. Section 5 presents model validation and major quantitative results (pending). Section 6 discusses the relationship of our paper and the existing literature. Section 7 concludes.

2 Empirical Analysis

In this section, we use Compustat data to examine the secular trends and cross-sectional relationships of intangible intensity, multinational production, and profit shifting in U.S. public firms. We also employ an event-study design to study how engaging in multinational production and profit shifting affect intangible intensity.

2.1 Data

Our primary dataset is the Compustat North America database, which includes all publicly listed firms from 1995 to 2023. We utilize the Annual Fundamentals dataset to extract consolidated, firm-level balance sheet and income statement information, including total revenue, cost of goods sold, research and development (R&D) expenditure, selling, general and administrative (SG&A) expense, as well as tangible and intangible assets. We only use the domestic version of the dataset, which has been standardized across firms and years for comparability. We exclude firm-year observations with missing industry codes and those with non-positive or missing values of total revenue, cost of goods sold, or SG&A expenses.

Under the Generally Accepted Accounting Principles (GAAP), reported intangible assets in the balance sheet reflect only externally acquired assets, as internally incurred expenses on intangibles cannot be capitalized. To measure firm-level intangible assets properly, we follow [Peters and Taylor \(2017\)](#) and construct a measure of internal intangible assets by capitalizing R&D expenditures using the perpetual-inventory method.¹ We then combine this internal measure with the balance-sheet measure to form the total intangible assets at the firm level.

¹We also create a version of internal intangible capital by additionally capitalizing 30% of the SG&A expenses as in [Eisfeldt and Papanikolaou \(2013\)](#). The results using this alternative measure are similar.

To measure the extent of multinational production, we use a linked database combining Compustat data with firm-level disclosures from the U.S. Securities and Exchange Commission (SEC), accessed through Wharton Research Data Services (WRDS). By the SEC rules, public firms in the U.S. are required to disclose the country of incorporation and ownership structure for subsidiaries in which they hold at least a 10% ownership stake.² We use this information to construct measures of each firm’s multinational expansion, including measures on the extensive margin (i.e., whether a firm operates internationally) and the intensive margin (i.e., the number of foreign countries).

2.2 Secular Trend

Figure 1 highlights three key trends in U.S. firms over the past three decades: the rise in intangible capital, the expansion of multinational production, and a growing tendency to engage in profit shifting. Panel (a) illustrates that average intangible intensity, measured as the ratio of intangible assets to total assets, has risen significantly from 22% in 1995 to 42% in 2023. A similar upward trajectory also holds true for the revenue-weighted average, which has increased from 16% in 1995 to 38% in 2023. This growing importance of non-rival intangible capital has been well-documented in the literature, such as [Corrado, Hulten, and Sichel \(2009\)](#) and [Crouzet, Eberly, Eisfeldt, and Papanikolaou \(2022\)](#).

Figure 2 shows that the secular increase in intangible intensity is primarily driven by within-industry changes rather than a compositional shift from low-intensity to high-intensity industries. Panel (a) reveals a significant rightward shift of the intangible intensity distribution from 1995 to 2023. Panel (b) shows that counterfactual series constructed using fixed revenue shares by industry in 1995 trace the actual intangible intensity series remarkably well. These two figures both suggest that the increase in intangible intensity has occurred across all industries of the economy, resulting in its increasing overall prominence.

Meanwhile, an increasing share of firms have engaged in multinational production and operate in more foreign countries. Panel (b) demonstrates significant growth in both the weighted and revenue-unweighted shares of multinational enterprises (MNEs), defined as firms reporting foreign subsidiaries. This trend reflects not only growth at the extensive margin—with more firms becoming multinational—but also at the intensive margin. Panel (c) further shows that the share of sales by the foreign subsidiaries by U.S. MNEs has also steadily increased. Over the last three decades, the unweighted (sales-weighted) foreign sales shares have grown from 10% (27%) to 27% (36%). In addition to declining FDI frictions, the rise of non-rival intangible capital is also a likely

²The relevant SEC filing is the Exhibit 21, including 10-K, 10-Q, 8-K, etc.

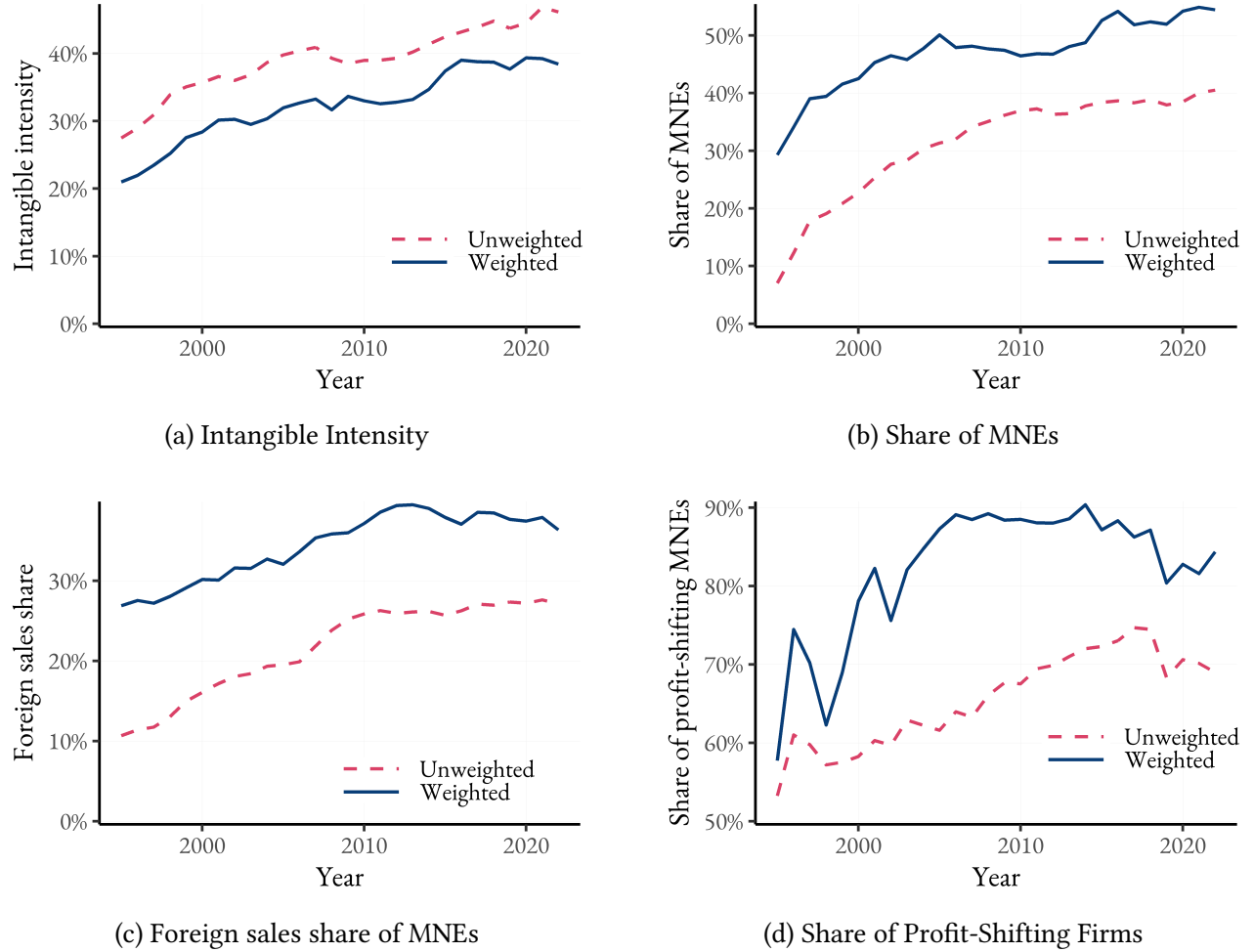


Figure 1: Secular Trends of Intangible Intensity, MNEs, and Profit Shifting: 1995 – 2023

Note: Intangible intensity is the ratio of intangible assets to total assets, where intangible assets include balance-sheet intangible assets and capitalized internal R&D following [Peters and Taylor \(2017\)](#), and total assets are defined as the sum of physical and intangible assets. Profit-shifting firms are those with foreign subsidiaries in any of the tax-haven countries identified by [Tørsløv, Wier, and Zucman \(2022\)](#).

cause to the increased multinational production. We aim to disentangle these forces underneath the secular trends through the lens of our model.

Lastly, panel (d) highlights the rise in profit shifting, evidenced by the increasing share of MNEs operating foreign subsidiaries in the tax-haven countries identified by [Tørsløv et al. \(2022\)](#). Notably, the share increased from 55% in 1995 to 74% in 2017 and then dropped significantly after the implementation of the Tax Cuts and Jobs Act, which lowers MNEs' tax incentive of profit shifting. Since larger firms are more likely to engage in profit shifting to lower global tax liabilities ([Delis, Delis, Laeven, and Ongena, 2024](#)), the revenue-weighted share not only is larger but also has expe-

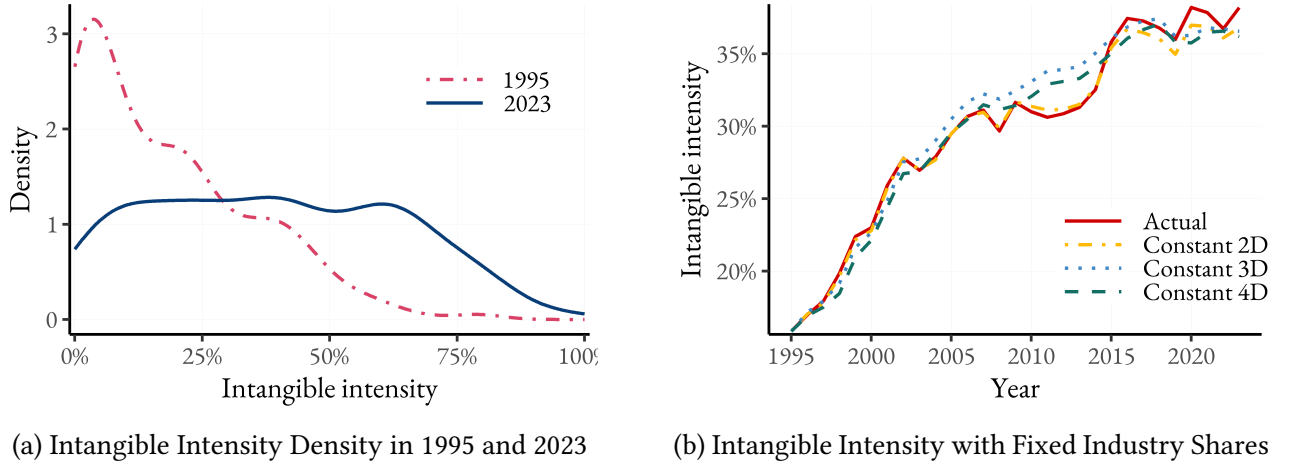


Figure 2: Intangible Intensity Across Industries Over Time

Note: Panel (a) shows the kernel density plot of intangible intensity aggregated to 4-digit NAICS industries in 1995 and 2023 using firm revenue weights. Panel (b) shows the counterfactual intangible intensity series where the revenue shares at different digit-levels of industry are held fixed at their 1995 levels. For more granular industry definitions, such as 4-digit NAICS industries, we focus only on NAICS industries present throughout the sample period.

rienced a greater increase. The co-movement of aggregate trends in intangible intensity and profit shifting aligns with the theory proposed in [Dyrda et al. \(2024a\)](#), in which MNEs exploit transfer pricing strategies of intangible assets to shift profits to the tax haven.

2.3 Cross-sectional relationship

We now examine the cross-sectional relationships between intangible intensity, multinational production, and profit shifting. Table 1 reports regressions of intangible intensity on measures of multinational production (MNE status and number of foreign countries) and a profit-shifting indicator. To examine how these relationships evolve over time, we interact the explanatory variables with a post-2005 dummy. All regressions control for firm size and include year fixed effects.

The results in Table 1 reveal that MNEs, particularly those operating in multiple countries and engaging in profit shifting, exhibit significantly higher intangible capital intensities. Specifically, during 1995-2005, the intangible intensity of MNEs is 11.1 percentage point higher than non-MNEs. Additionally, a 10% increase in the number of foreign countries in which a MNE operates is associated with a 0.3 percentage point increase in intangible intensity. Furthermore, profit-shifting MNEs on average had 10.1 percentage point higher intangible intensity compared to non-profit-shifting firms. These positive relationships are significantly more pronounced after 2005, as captured by the positive and statistically significant estimates of the interaction terms.

Table 1: Intangible Intensity, Multinational Production, and Profit Shifting

Dependent variable:	Intangible intensity		
	(1)	(2)	(3)
MNE indicator	0.111*** (0.005)		
MNE indicator x Post 2005	0.029*** (0.007)		
Log foreign countries of operation		0.045*** (0.001)	
Log foreign countries x Post 2005		0.014*** (0.002)	
Profit-shifting indicator			0.101*** (0.006)
Profit-shifting indicator x Post 2005			0.050*** (0.009)
Observations	147,360	147,360	147,360
Year FEs	Y	Y	Y
Control for log revenue	Y	Y	Y

Note: All three regressions are weighted by firm revenue and control for both year fixed-effects and log firm revenues. Standard errors are clustered at the year-level. $p < 0.10$ (*), $p < 0.05$ (**), $p < 0.01$ (***).

We note that the estimated cross-sectional relationship may be driven by many unobserved firm characteristics, such as productivity and tradability of the product. They are also subject to reverse causality – namely, high intangible intensity firms are more likely to open more foreign subsidiaries and engage in profit shifting. Therefore, we view these results as suggestive evidence and will later apply the quantitative model to shed light on the mechanisms.

2.4 Event study

We employ the interaction-weighted (IW) estimator, as introduced in [Sun and Abraham \(2021\)](#), to estimate the dynamic treatment effects of MNE status or profit-shifting on intangible intensity for a balanced panel of firms spanning 1995 to 2023. Specifically, we focus on the impact of a firm first attaining MNE status or initially engaging in profit-shifting.³ In the context of staggered treat-

³This represents an *absorbing* treatment, where $D_{js} \leq D_{jt}$ for $s < t$.

ment adoption, the IW estimator offers significant advantages. It mitigates contamination from treatment effects in other periods and accounts for treatment effect heterogeneity, an issue when applying the standard two-way fixed-effects (TWFE) event-study approach. To address variations in the duration that firms remain MNEs or engage in profit-shifting, we estimate treatment effects separately for cohorts of firms based on their initiation year of treatment. Specifically, we run the following regression

$$\text{Intangible intensity}_{jt} = \gamma_j + \delta_t + \sum_{e \notin C} \sum_{\tau \neq -1} \beta_{e,\tau} \cdot \mathbb{1}_{\{E_j=e\}} \cdot \mathbb{1}_{\{t-E_i=\tau\}} + \varepsilon_{jt}, \quad (1)$$

where E_j is the year when firm j becomes an MNE or begins profit shifting, $\beta_{e,\tau}$ is the cohort-specific average treatment effect on the treated (CATT) τ periods from initial treatment for the cohort of firms first treated in year e . We use firms that never become multinational enterprises (MNEs) or engage in profit-shifting as the control group C . We include all relative time indicators τ , excluding only the indicator for the period before treatment ($\tau = -1$) to address multicollinearity. The dynamic treatment effect estimate for each relative time period is given by the average of $\hat{\beta}_{e,\tau}$, weighted by the proportion of each cohort e in the relative time period τ .

We test the parallel trend assumption by including the leads of the treatment. The concern with a violation of the parallel trends assumption for this estimator centers on the timing of treatment and its relation to the evolution of intangible intensity had the firm never become an MNE or engaged in profit-shifting. We include firm revenue as a co-variate to control for the expansionary effect of the treated firms. In addition, firms which became an MNE or engaged in profit-shifting in earlier years plausibly could have had a different counterfactual evolution of intangible intensity, relative to firms which became MNEs or engaged in profit-shifting in later years. The IW estimator we apply addresses this concern. The results shown in Figure 3 indicate no statistically significant difference in the pre-trend for treated firms relative to firms which never become an MNE or engage in profit-shifting, which validates the identification assumption.

Figure 3 shows that both becoming an MNE and initiating profit-shifting activities are associated with sustained increases in intangible intensity. Intangible intensity increases by around 3 percentage point a year after the firm first produces abroad, with the effect increasing to 6 percentage points two years after treatment. Similarly, intangible intensity increases by around 3 percentage points two years after a firm begins profit shifting and it persists at this level over time. The lagged responses in intangible intensity suggest the presence of adjustment costs in intangible investment.

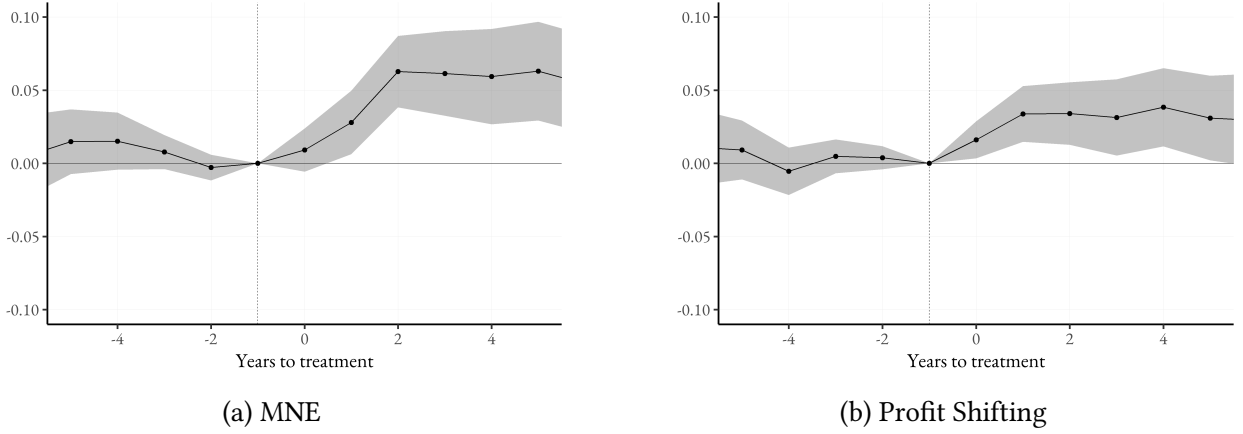


Figure 3: Event Study

Note: The figures display the estimates of β_τ in equation (1) for five relative time periods before and after the treatment, which is defined as either becoming an MNE or establishing a subsidiary in a tax-haven country. Standard errors are clustered at the firm and year level and the shaded area represents the 95%-level confidence interval of the β_τ estimates.

3 Quantitative Theory

To analyze the structural forces driving the rise of intangible capital, multinational expansion, and profit shifting, we develop a dynamic general equilibrium model that explicitly accounts for firms' investment, production, and tax optimization decisions. The model features heterogeneous firms that endogenously choose their level of intangible investment, foreign expansion, and profit reallocation across jurisdictions. The framework integrates households, final and intermediate goods producers, and governments across multiple regions, allowing us to assess the macroeconomic implications of technological change, globalization, and corporate taxation policies. In this section, we outline the key building blocks of the model, starting with household behavior and production technologies before detailing firms' decisions regarding intangible investment, multinational expansion, and tax-motivated profit shifting.

3.1 Households.

Each region i has a representative household with preferences over sequences of consumption, $\{C_{it}\}_{t=0}^\infty$, and labor supply, $\{L_{it}\}_{t=0}^\infty$, given by

$$\sum_{t=0}^{\infty} \beta^t \left[\log \left(\frac{C_{it}}{N_i} \right) + \psi_i \log \left(1 - \frac{L_{it}}{N_i} \right) \right]. \quad (2)$$

Households choose consumption, labor supply, tangible investment, $\{X_{it}\}_{t=0}^{\infty}$, and internationally-traded bonds, $\{B_{it+1}\}_{t=0}^{\infty}$ to maximize utility subject to a sequence of budget constraints,

$$P_{it}[C_{it} + K_{it+1} - (1 - \delta)K_{it}] + P_{bt}B_{it+1} = (1 - \tau_{ilt})W_{it}L_{it} + R_{it}K_{it} + B_{it} + D_{it} + T_{it}, \quad (3)$$

taking the wage, W_{it} , the labor income tax rate, τ_{ilt} , the rental rate, R_{it} , the bond price, P_{bt} , dividends, D_{it} , and lump-sum transfers T_{it} as given.

3.2 Technology

Final goods. Each region has a nontradable final good. Let Y_{it} denote gross output of the final goods. This can be used for consumption C_{it} , investment I_{it} , government consumption G_{it} or as materials X_{it} , so that

$$Y_{it} = C_{it} + I_{it} + G_{it} + X_{it} \quad (4)$$

The use of the final goods as materials gives the model a simple 'roundabout' production structure as for example in [Edmond, Midrigan, and Xu \(2023\)](#).

The final good Y_{it} is produced by perfectly competitive firms using inputs $q_{jit}(\omega)$ from a intermediate firms operating in i , i.e.

$$Y_{it} = \left[\sum_{j=1}^J \int_{\Omega_{jit}} q_{jit}(\omega)^{\frac{\varrho-1}{\varrho}} d\omega \right]^{\frac{\varrho}{\varrho-1}}, \quad (5)$$

where $\varrho > 1$ is the elasticity of substitution across varieties. Let $p_{jit}(\omega)$ denote the price of variety ω produced in i and sold in j . Then the final good price index in i is then given by

$$P_{it} = \left[\sum_{j=1}^J \int_{\Omega_{jit}} p_{jit}(\omega)^{1-\varrho} d\omega \right]^{\frac{1}{1-\varrho}}. \quad (6)$$

Hence the inverse demand for intermediate goods is given by

$$p_{jit}(\omega) = P_{it} \left(\frac{q_{jit}(\omega)}{Y_{it}} \right)^{-\frac{1}{\varrho}} \quad (7)$$

Intermediate Goods. Intermediate goods firms produce gross output according to the following technology.

$$y_{ijt}(\omega) = \sigma_{ij} A_j a \left(\zeta^{\frac{1}{\eta}} v_{it}(\omega)^{\frac{\eta-1}{\eta}} + (1 - \zeta)^{\frac{1}{\eta}} x_{it}(\omega)^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}} \quad (8)$$

where v_{it} is the firm's value-added, a composite of tangible capital, k_{ijt} , intangible capital, z_{it} , and labor l_{ijt} . Labor and tangible capital are rival, whereas intangible capital is nonrival within the scope of the firm.

$$v_{ijt}(\omega) = z_{it}(\omega)^\phi k_{ijt}(\omega)^\alpha l_{ijt}(\omega)^{1-\alpha-\phi}. \quad (9)$$

The parameters of this technology are as follows. A_j is region j 's aggregate total factor productivity, ϕ is the share of intangible capital and α is the share of tangible capital. $\sigma_{ij} \in [0, 1]$ represents a technological FDI barrier as in [McGrattan and Prescott \(2010\)](#); we assume no barriers in domestic production, i.e., $\sigma_{ii} = 1$.

Intangible Capital. Intangible capital is produced by an intermediate firm ω using labor as input and using the following technology:

$$z_{it}(\omega) = A_i a(\omega) l_{it}^z(\omega) \quad (10)$$

Once intangible capital is produced Intangible capital is nonrival within the firm, i.e. once developed it is Once the firm produces a stock $z_{it}(\omega)$ at home it allocates $\{z_j\}_{j \in J_F \cup \{i\}}$ across locations subject to

$$z_{it}(\omega) \geq \left(\sum_{j \in J_F \cup \{i\}} z_{ijt}(\omega)^{\frac{1}{1-\theta}} \right)^{1-\theta}. \quad (11)$$

Note that $\theta \in [0, 1]$ controls the degree of nonrivalry of intangible capital. If $\theta = 0$, then $z_{it}(\omega) = \sum_j z_{ijt}(\omega)$ and intangible capital is fully rival. Contrary, if $\theta = 1$, then $z_{it}(\omega) = \max_j z_{ijt}(\omega)$ and intangible capital is fully nonrival.

Trade and FDI. Intermediate can sell their products for free in the domestic market but accessing foreign markets is costly. Firms from region i pay a fixed cost κ_i^X to export and a fixed cost κ_i^F to produce locally in a foreign region. These costs are paid on a per-destination basis and are denominated in units of the home country's labor. There is also a variable exporting cost ξ_{ij} modeled as an iceberg cost. We denote a firm's set of export destinations by $J_{it}^X(\omega) \subseteq I \setminus \{i\}$ and its

set of FDI destinations by $J_{it}^F(\omega) \subseteq I \setminus \{i\}$. Given these sets, the firm's resource constraints are:

$$y_{iit}(\omega) = q_{iit}(\omega) + \sum_{j \in J_{it}^X(\omega)} \xi_{ij} q_{ijt}^X(\omega) \quad (12)$$

$$y_{ijt}(\omega) = q_{ijt}(\omega), \quad \forall j \in J_{it}^F(\omega) \quad (13)$$

where $q_{ijt}^X(\omega)$ and $q_{ijt}(\omega)$ represent exports and locally-produced products, respectively.

3.3 Firms: Profit Shifting

Multinationals' foreign affiliates are required to pay licensing fees for using intangible capital, adhering to transfer pricing rules. The licensing fee for a subsidiary in region j is given by $\vartheta_{ij}(z)z$, where $\vartheta_{ij}(z) = \phi p_{ij} y_{ij}/z$, representing the marginal revenue product of intangible capital under the "arm's length" rule. The total licensing fees across the conglomerate are $\nu_i(z)z = \sum_{j \in J_F \cup \{i\}} \vartheta_{ij}(z)z$, capturing the full value of the MNE's intangible capital, including fees for the parent corporation's use.

Multinationals based in high-tax regions can shift licensing income to low-tax regions or tax havens by transferring ownership of intangible capital to affiliates in these regions, provided fixed costs for establishing the affiliates are paid. If a fraction λ_{LT} is sold to a low-tax affiliate and a fraction λ_{TH} to a tax-haven affiliate, the LT affiliate collects $\lambda_{LT} \sum_{j \in J_F \cup \{i\} \setminus \{LT\}} \vartheta_{ij}(z)z$, the TH affiliate collects $\lambda_{TH} \sum_{j \in J_F \cup \{i\}} \vartheta_{ij}(z)z$, and the domestic parent collects the remaining $(1 - \lambda_{LT} - \lambda_{TH}) \sum_{j \in J_F} \vartheta_{ij}(z)z$. Multinationals in low-tax regions do not engage in profit shifting.

The sale of intangible capital from a domestic parent in region i to an affiliate in region $j \in \{LT, TH\}$ occurs at a markdown $\mu_{ij} < 1$ below market value $\nu_i(z)$. This markdown incentivizes profit shifting by creating additional profits when assets are transferred at reduced prices. If sales occurred at fair market value ($\mu_{ij} = 1$), profit shifting would have no financial impact. The markdown is essential for motivating profit shifting.

However, shifting profits involves costs. The cost of transferring fractions λ_{LT} and λ_{TH} to low-tax regions is denoted by $\mathcal{C}_i(\lambda_{LT}, \lambda_{TH})$, which satisfies the following assumptions:

Assumption 1. Let $\boldsymbol{\lambda} = [\lambda_{LT}, \lambda_{TH}]$ and

$$\Gamma = \{\boldsymbol{\lambda} \in [0, 1]^2 : \lambda_{LT} + \lambda_{TH} \leq 1\}. \quad (14)$$

The cost function $\mathcal{C}_i : \Gamma \rightarrow \mathbb{R}_+$ has the following properties:

1. *At least twice continuously differentiable.*
2. $\mathcal{C}_i^j \equiv \frac{\partial \mathcal{C}_i}{\partial \lambda_j} > 0$ for $j \in \{LT, TH\}$.
3. $\mathcal{C}_i^{jj} \equiv \frac{\partial^2 \mathcal{C}_i}{\partial \lambda_j^2} > 0$ for $j \in \{LT, TH\}$.
4. $\mathcal{C}_i(\mathbf{0}) = 0$. Additionally, for every sequence $\{\boldsymbol{\lambda}_n\} \subseteq \Gamma$ with $\boldsymbol{\lambda}_n \rightarrow \boldsymbol{\lambda}^*$ and $\lambda_{LT}^* + \lambda_{TH}^* = 1$, it holds that $\lim_{n \rightarrow \infty} \mathcal{C}_i^j(\boldsymbol{\lambda}_n) = \infty$ for $j \in \{LT, TH\}$.

The first assumption ensures smoothness for mathematical convenience. The second implies that costs increase with the amount transferred. The third ensures marginal costs grow with additional transfers. The fourth establishes that no costs occur without transfers, and transferring the full stock is prohibitively costly.

The function $\mathcal{C}_i(\lambda)$ captures direct costs (e.g., legal and consulting fees) and indirect risks (e.g., penalties for non-compliance), as discussed in [Allingham and Sandmo \(1972\)](#); [Rotberg and Steinberg \(2022\)](#).

3.4 Firms: Profit Maximization

We present here in detail a profit maximization problem for firms in our model. For the sake of notational brevity we skip the dependence of all objects on variety ω and time t . For the expository reasons we split the problem into the intensive margin and extensive margin. In the former firms choose allocation of inputs, quantities and share of shifted rights to intellectual property. In the latter firms choose export and FDI locations.

Intensive Margin. It is useful to first define pre-tax operating profits, π_i^D , for domestic (home) subsidiary

$$\pi_i^D(a; J_X) = p_{ii}(q_{ii})q_{ii} + \sum_{j \in J_X} p_{ij}(q_{ij}^X)q_{ij}^X - W_i(l_{ii} + l_i^z) - R_i(\delta + k_{ii}) - P_i x_{ii} \quad (15)$$

and pre-tax operating profits, π_i^D , for foreign subsidiary

$$\pi_i^F(a; J_F) = p_{ij}(q_{ij})q_{ij} - W_j l_{ij} - R_i(\delta + k_{ij}) - P_j x_{ij}. \quad (16)$$

These are simply revenues of input costs for each subsidiary of the MNE. We then can define global pre-tax profits of the MNE as follows:

$$\begin{aligned} \pi_i^G(a; J_X, J_F, J_{TH}) &= \max_{\mathcal{A}_i} \left\{ \pi_{ii} + \sum_{j \in J_F} (1 - \tau_j) \pi_{ij} + \pi_{iTH} - T_i^G \right\} \\ \text{s.t.} \quad & (7) - (14) \end{aligned} \quad (17)$$

where the choice set of the MNE is given by

$$\mathcal{A}_i = \left\{ q_{ii}, l_{ii}, k_{ii}, \{q_{ij}, l_{ij}, k_{ij}\}_{j \in J_F}, \{q_{ij}^X\}_{j \in J_X}, z, \{\lambda_{LT}, \lambda_{TH}\} \in \Gamma, \lambda_{LT} \leq \mathbb{1}_{\{LT \in J_F\}}, \lambda_{TH} \leq \mathbb{1}_{\{J_{TH}=1\}} \right\}$$

The firm can sell its intangible capital to either a low-tax region or tax haven only conditional on having established a subsidiary there, which is reflected in the two inequality constraints on λ_{LT} and λ_{TH} accordingly. The first term in (17) represents the pre-tax profits of the parent division, π_{ii} . The second term represents the pre-tax profits in foreign subsidiaries, $\pi_{i,j}$, potentially including low-tax region LT and taxed at corporate tax rates τ_j in j . The third represents the pre-tax profits of the tax-haven affiliate, $\pi_{i,TH}$. Finally, the last term T_i^G represents global tax liability of a MNE headquartered in i . We now discuss the components of (17) in more detail.

The first term in (18) represents domestic operating profits as defined in (15). The second term is the revenue from selling marked-down intangible capital to the two profit-shifting destinations. The third term represents the licensing fee income associated with intangible capital retained by the parent. The fourth term contains the associated licensing fees paid by the parent division on the transferred intangible capital. The fifth term shows the costs of profit shifting.

$$\begin{aligned} \pi_{ii} &= \pi_i^D(a; J_X) + \overbrace{(\mu_{iLT} \lambda_{LT} + \mu_{iTH} \lambda_{TH}) \nu_i(z) z}^{\text{Proceeds from selling } z} + \overbrace{\sum_{j \in J_F} (1 - \lambda_{LT} - \lambda_{TH}) \vartheta_{ij}(z) z}^{\text{Licensing fee receipts}} \\ &\quad - \overbrace{(\lambda_{LT} + \lambda_{TH}) \vartheta_{ii}(z) z}^{\text{Licensing fee payments}} - \overbrace{W_i \mathcal{C}_i(\lambda_{LT}, \lambda_{TH}) \nu_i(z) z}^{\text{Cost of transferring } z}. \end{aligned} \quad (18)$$

Taxable profit of a subsidiary in the low tax region is given by

$$\begin{aligned} \pi_{iLT} &= \pi_{iLT}^F(a; J_F) + \overbrace{\sum_{j \in J_F \cup \{i\} \setminus \{LT\}} \lambda_{LT} \vartheta_{ij}(z) z}^{\text{Licensing fee receipts}} - \underbrace{\mu_{iLT} \lambda_{LT} \nu_i(z) z}_{\text{Cost of buying } z} - \underbrace{(1 - \lambda_{LT}) \vartheta_{iLT}(z) z}_{\text{Licensing fee payment}} \end{aligned} \quad (19)$$

The first term in (19) is operating profit as defined in (16). The second term represents the licensing fees collected on the fraction of intangible capital transferred to the LT region, λ_{LT} . The third term represents the marked-down cost of purchasing this intangible capital from the parent division. The last term represents the licensing fee the affiliate in this region pays on the fraction of intangible capital that it does not purchase, $1 - \lambda_{LT}$.

Taxable profits of a subsidiary in another high-tax region j are

$$\pi_{ij} = \pi_{ij}^F(a; J_F) - \underbrace{\vartheta_{ij}(z)z}_{\text{Licensing fee}}. \quad (20)$$

The two terms are the operating profit defined in (16) and the licensing fees paid for the use of intangible capital, respectively. Finally, taxable profits of a subsidiary in the unproductive tax haven, are defined as

$$\pi_{iTH} = \underbrace{\sum_{j \in J_F \cup \{i\}} \lambda_{TH} \vartheta_{ij}(z)z}_{\text{Licensing fee receipts}} - \underbrace{\mu_{iTH} \lambda_{TH} \nu_i(z)z}_{\text{Cost of buying } z}. \quad (21)$$

The first term in (21) represents the licensing fees collected on the fraction of intangible capital transferred to the TH region, λ_{TH} . The second term represents the marked-down cost of purchasing this capital. Finally, global tax liability is given by

$$T_i^G = \sum_{j \in J_F \cup i} \tau_j (\pi_{ij} + R_j k_{ij}) + \tau_{TH} \pi_{TH} \quad (22)$$

Note that, consistent with standard accounting practices, depreciation on tangible capital is tax-deductible but rental costs are not, thus we have to adjust the pre-tax profits for that in (22).

Extensive margin. MNE chooses export destinations and foreign affiliates by solving the following problem of the firm

$$d_i(a) = \max_{J_X, J_F, J_{TH}} \left\{ \pi_i^G(a; J_X, J_F, J_{TH}) - (1 - \tau_i) W_i \underbrace{\left(\sum_{j \in J_X} \kappa_{ij}^X + \sum_{j \in J_F} \kappa_{ij}^F + \kappa_i^{TH} \mathbb{1}_{\{J_{TH}=1\}} \right)}_{\text{Fixed costs}} \right\} \quad (23)$$

where $\pi_i^G(a; J_X, J_F, J_{TH})$ are the after-tax global profits defined in (17) from which the net-of-taxes fixed costs are subtracted.

3.5 Aggregation and accounting measures

Several national and international accounting measures are required to close the model and compare it to the data. Here, we revert to expressing firms' choices as functions of their varieties (ω) for notational brevity.

Gross domestic product. Nominal GDP is the total value added produced in a given region:

$$GDP_i = \sum_{j=1}^I \int_{\omega \in \Omega_j, i \in J_F(\omega)} p_{ji}(\omega) v_{ji}(\omega) d\omega. \quad (24)$$

We compute real GDP by deflating by the consumer price index P_i defined in (6).

Goods trade. Aggregate goods trade flows are given by

$$EX_i^G = \sum_{j \neq i} \int_{\Omega_i} p_{ij}^X(\omega) (1 + \xi_{ij}) q_{ij}^X(\omega) d\omega, \quad (25)$$

$$IM_i^G = \sum_{j \neq i} \int_{\Omega_j} p_{ji}^X(\omega) (1 + \xi_{ji}) q_{ji}^X(\omega) d\omega. \quad (26)$$

Services trade. Consistent with [Guvenen, Mataloni, Rassier, and Ruhl \(2022\)](#) and [Accoto, Federico, and Oddo \(2021\)](#), intangible capital licensing fees enter the national accounts as exports or

imports of intellectual property services.⁴ High-tax regions' services trade flows are given by

$$EX_i^S = \sum_{j \neq i} \int_{\Omega_i} [1 - \lambda_{LT}(\omega) - \lambda_{TH}(\omega)] \vartheta_{ij}(\omega) z(\omega) d\omega \quad (27)$$

$$+ \int_{\Omega_i} [\mu_{i,LT} \lambda_{LT}(\omega) + \mu_{i,TH} \lambda_{TH}(\omega)] \nu_i(\omega) z(\omega) d\omega$$

$$IM_i^S = \sum_{j \neq i} \int_{\Omega_i} [\lambda_{LT}(\omega) + \lambda_{TH}(\omega)] \vartheta_{ij}(\omega) z(\omega) d\omega \quad (28)$$

$$+ \sum_{j \neq i} \int_{\Omega_j} \vartheta_{ji}(\omega) z(\omega) d\omega.$$

The low-tax region's services trade flows are

$$EX_{LT}^S = \sum_{j \neq LT} \int_{\Omega_{LT}} \vartheta_{LT,j}(\omega) z(\omega) d\omega + \sum_{j \neq LT} \int_{\Omega_j} \lambda_{LT}(\omega) \sum_{k \neq LT} \vartheta_{j,k}(\omega) z(\omega) d\omega, \quad (29)$$

$$IM_{LT}^S = \sum_{j \neq LT} \int_{\Omega_j} [1 - \lambda_{LT}(\omega)] \vartheta_{j,LT}(\omega) z(\omega) d\omega + \sum_{j \neq LT} \int_{\Omega_j} \mu_{j,LT} \lambda_{LT}(\omega) \nu_j(\omega) z(\omega) d\omega. \quad (30)$$

The tax haven's services trade flows are

$$EX_{TH}^S = \sum_{j \neq TH} \int_{\Omega_j} \lambda_{TH}(\omega) \nu_j(\omega) z(\omega) d\omega, \quad (31)$$

$$IM_{TH}^S = \sum_{j \neq TH} \int_{\Omega_j} \mu_{j,TH} \lambda_{TH}(\omega) \nu_j(\omega) z(\omega) d\omega. \quad (32)$$

Net factor receipts and payments. Net factor receipts from (payments to) foreigners are the sum total of the dividends paid by foreign subsidiaries of domestic multinationals (domestic subsidiaries of foreign multinationals):

$$NFR_i = \sum_{j \neq i} \int_{\Omega_i} (1 - \tau_j) \pi_{ij}(\omega) d\omega, \quad (33)$$

$$NFP_i = \sum_{j \neq i} \int_{\Omega_j} (1 - \tau_i) \pi_{ji}(\omega) d\omega. \quad (34)$$

⁴For simplicity, we include marked-down transfers of intangible capital ownership as services trade as well as licensing fees. In reality, these transfers may actually enter the financial account depending on how they are structured. However, due to the balance of payments identity this does not affect the way these transfers affect the overall balance of payments. Our accounting allows us to avoid defining the financial account altogether.

3.6 Market clearing and equilibrium

In a general equilibrium of our model, the labor market must clear, the government's budget constraint must be satisfied, and the balance of payments must hold in each productive region.

Labor market. Labor demand comes from four sources: production of intermediate goods; production of intangible capital; fixed costs of exporting and setting up foreign affiliates; and the costs of transferring intangible capital. The labor market clearing condition can be written as

$$\begin{aligned}
 L_{it} = & \underbrace{\sum_{j=1}^I \int_{\Omega_j} \ell_{ji}(\omega) d\omega}_{\text{goods production}} + \underbrace{\int_{\Omega_i} l_i^z d\omega}_{\text{z production}} + \underbrace{\int_{\Omega_i} \left(\sum_{j \in J_X(\omega)} \kappa_i^X + \sum_{j \in J_F(\omega)} \kappa_i^F + \kappa_i^{TH} \mathbb{1}_{\{J_{TH}(\omega)=1\}} \right) d\omega}_{\text{fixed costs}} \\
 & + \underbrace{\int_{\Omega_i} \mathcal{C}_i(\lambda_{LT}, \lambda_{TH}) \nu(\omega) z(\omega) d\omega}_{\text{costs of shifting } z}. \tag{35}
 \end{aligned}$$

Capital market. The capital market clearing condition is

$$K_{it} = \sum_{j=1}^I \int_{\Omega_j} k_{jit}(\omega) d\omega. \tag{36}$$

Government budget constraint. We assume that revenue from corporate and labor income taxation is rebated lump-sum to households.⁵ Lump-sum transfers are given by

$$T_i = \tau_i \sum_{j=1}^I \int_{\Omega_j} \pi_{ji}(\omega) d\omega + \tau_{i\ell} W_{it} L_{it}. \tag{37}$$

Balance of payments. The balance of payments requires that each region's current account must be zero:

$$EX_i^G + EX_i^S - IM_i^G - IM_i^S + NFR_i - NFP_i = 0. \tag{38}$$

⁵With a fixed supply of labor, this assumption is without loss of generality. We have also analyzed a version of the model with endogenous labor supply. In this model, the results are similar with lump-sum transfers and when labor income taxes adjust to clear the government's budget constraint.

Note that several things happen to the balance of payments when a firm shifts profits away from its home region. First, that region's services trade balance worsens: the firm receives fewer licensing fees from its foreign subsidiaries and makes more licensing payments. Second, net factor receipts rise: the firm's profits in the tax haven (or low-tax region) rise, and these increased profits are ultimately rebated back to the home country. These two effects offset one another, but not completely: some of the shifted profits are taxed and therefore remain in the tax haven and/or low-tax region. Thus, the net effect is that the current account worsens. The reduction in the services trade balance and the increase in net factor income are consistent with the accounting of [Guvenen et al. \(2022\)](#). The net negative effect on the balance of payments is consistent with the findings of [Hebous, Klemm, and Wu \(2021\)](#). To regain equilibrium, that trade balance and/or net factor income balance must improve, which shows up in our model as a real exchange rate depreciation.

Competitive equilibrium. Given a set of parameters, an equilibrium in our model is defined as a set of aggregate prices and quantities $\{W_i, P_i, C_i\}$ and a set of firm decision rules $\{J_X(\omega), J_F(\omega), J_{TH}(\omega), z(\omega), \ell(\omega), \mathbf{q}(\omega), \lambda_{LT}(\omega), \lambda_{TH}(\omega)\}$ for each productive region $i \in I$ that satisfy the household's problem, the firm's problem, and the market-clearing conditions.

4 Theoretical Results

In this section, we develop a theoretical framework to analyze the model under consideration, aiming to illustrate the intricate nexus between intangible capital, multinational expansions, and profit shifting. To ensure analytical clarity and tractability, we make several simplifying assumptions. First, we abstract from tangible capital and materials, consolidating all rival factors into labor. Second, we focus on domestic production decisions, excluding the firm's exporting choices. Third, we assume that the fixed costs associated with establishing local production in foreign regions, denoted as κ_i^F , follow a Type-I Generalized Extreme Value distribution (Gumbel). Finally, our analysis concentrates on profit shifting to tax havens, explicitly abstracting from profit shifting to low-tax regions. We begin with setting up some necessary preliminaries, defining relevant objects and then move to the theoretical characterizations.

4.1 Preliminaries

Intangible Capital and Profit Shifting. The first lemma formalizes how firms allocate intangible capital when it is fully nonrival ($\theta = 1$). In this case, the equilibrium stock of intangible capital, z , depends on three key factors: (i) the after-tax marginal cost of intangible production, MC_{zi} , (ii) the relative size and productivity of foreign markets, captured by $\mathcal{P}_{ij}$ and $\mathcal{D}_j$, and (iii)

tax and profit-shifting incentives, reflected in $\Omega_{ij}(\lambda)$.

Lemma 2. *The allocation of intangible capital is given under fully nonrival case ($\theta = 1$) is given by*

$$z = \left[\left(\frac{\alpha}{(1 - \tau_i) MC_{zi}} \right) \sum_{j \in J_F \cup i} (\mathcal{P}_{ij} \times \mathcal{D}_j^\rho \times \Xi^\rho \times \Omega_{ij}(\lambda)^\rho)^{\frac{1}{1 + (\rho - 1)\alpha}} \right]^{1 + (\rho - 1)\alpha} \quad (39)$$

where

$$\begin{aligned} \mathcal{P}_{ij} &\equiv ((1 - \sigma_{ij}) a A_j)^{(\rho - 1)} \left(\frac{(1 - \alpha)}{W_j} \right)^{(1 - \alpha)(\rho - 1)} (1 - \tau_j)^{1 + \alpha(\rho - 1)} \\ \mathcal{D}_j &\equiv \left(Y_j^{\frac{1}{\rho}} P_j \right) \\ \Omega_{ij}(\lambda) &\equiv 1 + \alpha \left(\frac{(\tau_j - \tau_i) + (\tau_i - \tau_H)(1 - \mu)\lambda - (1 - \tau_i)W_i \mathcal{C}(\lambda)}{1 - \tau_j} \right) \\ \Xi &= \left(1 - \frac{1}{\rho} \right) \end{aligned}$$

The term $\mathcal{P}_{ij}$ captures the effect of foreign production costs and trade frictions on the demand for intangible capital. Countries with higher productivity (A_j) and lower wages (W_j) make intangible investment more attractive, while barriers to foreign direct investment (FDI), σ_{ij} , act as a friction. The term $\mathcal{D}_j$ represents demand-side factors, scaling with total output and the price index in each country.

Crucially, the expression $\Omega_{ij}(\lambda)$ shows how taxation interacts with intangible capital allocation. A firm's incentive to accumulate intangibles increases when tax differentials across jurisdictions are large ($\tau_j - \tau_i$) and when shifting profits to tax havens (τ_H) is feasible and inexpensive ($1 - \mu$). However, these benefits are offset by enforcement and compliance costs, $\mathcal{C}(\lambda)$, which limit the extent of profit shifting.

Relevant Definitions. First, let us define a global profit for MNE that has no foreign subsidiary in a productive region, $\pi_i^G(a, \emptyset)$, but has a subsidiary in a tax haven. Together with our simplifying assumption (17) becomes

$$\begin{aligned} \pi_i^G(a, \emptyset) &\equiv \max_{\mathcal{A}_i^\emptyset} (1 - \tau_i) [\Upsilon(q_{ii}) q_{ii} - W_i l_i - MC_{zi} z + \mu \lambda v_i(z) z - \lambda \vartheta_{ii}(q_{ii}, l_i, z) z - W_i \mathcal{C}(\lambda) v_i(z) z] \\ &\quad + (1 - \tau_H) [v_i(z) z \lambda (1 - \mu)] \end{aligned}$$

where $\mathcal{A}_i^\emptyset = \{z, q_{ii}, l_i, \lambda\}$. Then, let us define a global profit of the MNE that has a single subsidiary in region j , $\pi_i^G(a, 1)$. Together with our simplifying assumption (17) becomes in this case

$$\begin{aligned} \pi_i^G(a, 1) \equiv & \max_{\mathcal{A}_i^1} (1 - \tau_i) [\mathcal{T}(q_{ii}) q_{ii} - W_i l_i - MC_{zi} z + \mu \lambda v_i(z) z - \lambda \vartheta_{ii}(q_{ii}, l_i, z) z \\ & + (1 - \lambda) \sum_{j \in J_F} \vartheta_{ij}(q_{ij}, l_j, z) z - W_i C(\lambda) v_i(z) z] + (1 - \tau_H) [v_i(z) z \lambda (1 - \mu)] \\ & + (1 - \tau_j) [\mathcal{T}(q_{ij}) q_{ij} - W_j l_j - \vartheta_{ij}(q_{ij}, l_j, z) z] \end{aligned}$$

where $\mathcal{A}_i^1 = \{z, q_{ii}, l_i, q_{ij}, l_j, \lambda\}$. We now impose an assumption that the fixed costs of producing locally in a foreign region are distributed according to a Gumbell shock i.e. $\kappa_{ij}^F \sim \text{Gumbel}(0, \sigma_\kappa)$. The MNE would open a subsidiary

$$\kappa_{ij}^F < \pi_i^G(a, 1) - \pi_i^G(a, \emptyset) \quad (40)$$

and under assumption about Gumbell distribution this discrete decision can converted to a probabilistic one. This leads us to a definition of a *propensity to expand*, which can be formalized as follows.

Definition 1 (Expansion propensity). *Under the assumption $\kappa_{ij}^F \sim \text{Gumbel}(0, \sigma_\kappa)$ the probability that (40) holds can be expressed as:*

$$p_i(a) = \Pr(\kappa_{ij}^F < \pi_i^G(a, 1) - \pi_i^G(a, \emptyset)) = \frac{\exp\left\{\frac{\pi_i^G(a, 1) - \pi_i^G(a, \emptyset)}{\sigma_\kappa}\right\}}{1 + \exp\left\{\frac{\pi_i^G(a, 1) - \pi_i^G(a, \emptyset)}{\sigma_\kappa}\right\}}. \quad (41)$$

The second object of our interest is a model counterpart of *intangible intensity* we defined in the empirical section as the share of intangible investment to costs of goods sold. We have a similar definition in terms of model objects.

Definition 2 (Intangible intensity). *Let intangible intensity be the ratio of intangible capital to sum of all rival factors of production:*

$$s_i^z(a) = \frac{z_i}{\sum_{j \in J_F \cup i} l_{ij}}$$

Our third model object of interest captures the degree to which firms use their right to sell rights to intangible capital to move profits away from high tax jurisdictions to low tax jurisdictions. To

capture this we introduce a notion of shifted profits' share as follows.

Definition 3 (Shifted Profits' Share). Let π_i^{NS} be global profits a firm would have reported in region i if it did not shift profits, holding fixed all of its other policy functions.

$$ps_i = \frac{(\pi_i^{NS}(a) - \pi_i^G(a))}{\pi_i^G(a)} \quad (42)$$

Note that as $ps_i > 0$, this indicates that the firm has shifted profits away from region i .

4.2 Intangible Intensity and International Expansion.

Proposition 1 (Marginal Costs and Nonrivalry). Under the aggregator constraint (11) with $\theta \in [0, 1]$, and given the definitions of intangible intensity (s_i^z) and expansion propensity (p_i), the following hold:

1. **Negative Dependence on Marginal Cost MC_{zi} :**

For any $\theta \in [0, 1]$, both intangible intensity and expansion propensity decrease as the marginal cost of intangible production (MC_{zi}) increases:

$$\frac{\partial s_i^z(a, \theta)}{\partial MC_{zi}} < 0, \quad \frac{\partial p_i(a, \theta)}{\partial MC_{zi}} < 0.$$

2. **Sensitivity to Nonrivalry (θ):**

The elasticities of both $s_i^z(a, \theta)$ and $p_i(a, \theta)$ with respect to MC_{zi} grow in magnitude as θ increases:

$$|\varepsilon_{s_i^z}(\theta'')| > |\varepsilon_{s_i^z}(\theta')|, \quad |\varepsilon_{p_i}(\theta'')| > |\varepsilon_{p_i}(\theta')|,$$

for $0 \leq \theta' < \theta'' \leq 1$. Specifically:

- **Fully Rival ($\theta = 0$):** Intangible capital is perfectly rival ($z = \sum_j z_j$), yielding minimal elasticities.
- **Fully Nonrival ($\theta = 1$):** Intangible capital is perfectly nonrival ($z = \max_j z_j$), yielding maximal elasticities.

The proposition highlights the pivotal role of intangible nonrivalry in shaping multinational enterprises' (MNEs) decisions regarding intangible intensity and international expansion. When intangible capital is more nonrival (higher θ), it can be deployed across multiple operations without

significant depreciation in productivity. This scalability amplifies the sensitivity of firms' decisions to the costs of intangible production (MC_{zi}), as the returns to such investments are higher in more nonrival settings. Consequently, an increase in MC_{zi} significantly curtails intangible intensity (s_i^z) and the propensity to expand internationally (p_i) in industries where intangibles are highly scalable. Conversely, in industries where intangibles are more rival ($\theta \rightarrow 0$), their limited scalability dampens the impact of MC_{zi} , reducing firms' responsiveness to changes in intangible production costs.

Proposition 1's theoretical predictions align closely with the empirical patterns documented in Section 2. The model posits that reductions in the marginal cost of intangible production (MC_{zi}) increase intangible intensity (s_i^z) and expansion propensity (p_i), with these effects amplified by the degree of nonrivalry (θ). Empirically, the rise in intangible intensity from 22% to 42% over 1995–2023 (Figure 1a) coincides with technological advancements (e.g., cheaper software development, digitalization) that lowered MC_{zi} , particularly in nonrival-intensive sectors like technology. The cross-sectional evidence further supports this: MNEs and profit-shifting firms, which inherently leverage scalable intangibles (high θ), exhibit 8–9 percentage point higher intangible intensity than non-MNEs (Table 1). Additionally, the event study (Figure 3) shows that firms experience persistent increases in intangible intensity after becoming MNEs—a transition that enables exploiting nonrivalry, consistent with Proposition 1's mechanism. The within-industry growth in intangible capital (Figure 2a) further underscores the role of falling MC_{zi} and rising θ , as compositional shifts across industries cannot explain the trend. Together, the empirical evidence validates Proposition 1's core insight: nonrivalry amplifies firms' responsiveness to cost-driven intangible investment, a dynamic central to the rise of globalized, intangible-intensive MNEs.

Proposition 2 (FDI Barriers, Fixed Costs, and Nonrivalry). *Under the aggregator constraint (11) with $\theta \in [0, 1]$, and given the definitions of intangible intensity (s_i^z) and expansion propensity (p_i), the following hold:*

1. Negative Dependence on FDI Barriers and Fixed Costs:

For any $\theta \in [0, 1]$, the intangible intensity ($s_i^z(a, \theta)$) and the expansion propensity ($p_i(a, \theta)$) both decrease as the FDI barrier (σ_{ij}) or the fixed cost (κ_{ij}) increases:

$$\frac{\partial s_i^z(a, \theta)}{\partial \sigma_{ij}} < 0, \quad \frac{\partial s_i^z(a, \theta)}{\partial \kappa_{ij}} < 0, \quad \frac{\partial p_i(a, \theta)}{\partial \sigma_{ij}} < 0, \quad \frac{\partial p_i(a, \theta)}{\partial \kappa_{ij}} < 0.$$

Equivalently, reducing FDI barriers ($\sigma_{ij} \downarrow$) or fixed costs ($\kappa_{ij} \downarrow$) increases $s_i^z(a, \theta)$ and $p_i(a, \theta)$.

2. *Elasticity and Sensitivity to Nonrivalry (θ):*

The elasticities of $s_i^z(a, \theta)$ and $p_i(a, \theta)$ with respect to FDI barriers (σ_{ij}) and fixed costs (κ_{ij}) increase as θ rises:

$$\begin{aligned} |\varepsilon_{s_i^z, \sigma}(\theta'')| &> |\varepsilon_{s_i^z, \sigma}(\theta')|, & |\varepsilon_{p_i, \sigma}(\theta'')| &> |\varepsilon_{p_i, \sigma}(\theta')|, \\ |\varepsilon_{s_i^z, \kappa}(\theta'')| &> |\varepsilon_{s_i^z, \kappa}(\theta')|, & |\varepsilon_{p_i, \kappa}(\theta'')| &> |\varepsilon_{p_i, \kappa}(\theta')|, \end{aligned}$$

for $0 \leq \theta' < \theta'' \leq 1$. Specifically:

- **Fully Rival ($\theta = 0$):** *Intangible capital is perfectly rival ($z = \sum_j z_j$), yielding the smallest elasticities.*
- **Fully Nonrival ($\theta = 1$):** *Intangible capital is perfectly nonrival ($z = \max_j z_j$), yielding the largest elasticities.*

This proposition highlights the economic forces shaping the behavior of multinational enterprises (MNEs) in response to globalization, represented by reduced barriers to foreign direct investment (FDI) and lower fixed costs of establishing foreign subsidiaries. The first result establishes that both the intangible intensity of firms and their propensity to expand internationally decline as FDI barriers or fixed costs increase. These results reflect the critical role of globalization in enabling firms to leverage intangible capital across borders more effectively. By lowering the costs of accessing foreign markets, globalization enhances firms' ability to scale their intangible assets, driving greater intangible intensity and more frequent multinational expansions. Conversely, higher barriers or fixed costs act as frictions, constraining firms' ability to capitalize on the nonrivalry of intangible capital and reducing the benefits of international expansion.

The second result emphasizes the increasing importance of nonrivalry (θ) in determining firms' responsiveness to changes in globalization forces. As θ rises, intangible capital becomes more scalable, allowing firms to derive greater returns from deploying it across multiple markets. Consequently, the elasticities of intangible intensity and expansion propensity with respect to FDI barriers and fixed costs grow with θ . This implies that industries characterized by highly nonrival intangibles (e.g., technology, software) are more sensitive to globalization, exhibiting sharper increases in multinational activity and intangible intensity when barriers and fixed costs decline. Conversely, industries with more rival intangibles (e.g., manufacturing) are less responsive. These insights suggest that globalization disproportionately benefits sectors with high intangible nonrivalry.

Proposition 2's theoretical predictions are strongly supported by the empirical evidence in Section 2. The model asserts that reductions in FDI barriers ($\sigma_{ij} \downarrow$) and fixed costs ($\kappa_{ij} \downarrow$) increase intangible intensity (s_i^z) and expansion propensity (p_i), with these effects magnified in industries characterized by high nonrivalry (θ). Empirically, the expansion of multinational production—evidenced by the rise in the share of MNEs from 16% to 38% (Figure 1b) and the growth in the average number of countries per MNE from 8 to 12 (Panel c)—aligns with declining FDI barriers and reductions in fixed costs. Cross-sectional regressions (Table 1) reveal that MNEs operating in more foreign countries exhibit significantly higher intangible intensity, consistent with the mechanism whereby lower σ_{ij} and κ_{ij} enable firms to scale nonrival intangibles (e.g., software, IP). The event-study results (Figure 3) further validate this: firms that become MNEs experience sustained increases in intangible intensity, reflecting the model's predicted linkage between foreign expansion ($p_i \uparrow$) and intangible accumulation. The rise in profit shifting (Figure 1d), particularly in high- θ sectors, underscores how lower barriers and scalable intangibles jointly incentivize profit reallocation. Finally, the within-industry growth in intangible capital (Figure 2b) and the rightward shift in its distribution (Figure 2a) cannot be explained by compositional changes, further corroborating Proposition 2's focus on falling σ_{ij} and κ_{ij} as drivers of intangible intensity. Together, the evidence confirms that globalization forces and nonrivalry interact to shape the rise of intangible-intensive MNEs.

Definition 4. The *combined elasticity* of x with respect to σ_{ij} and MC_{zi} is defined as:

$$\varepsilon_{x,(\sigma_{ij} \cdot MC_{zi})} = \underbrace{\frac{\partial \ln x}{\partial \ln \sigma_{ij}}}_{\text{Elasticity to FDI barriers}} + \underbrace{\frac{\partial \ln x}{\partial \ln MC_{zi}}}_{\text{Elasticity to costs}} + \underbrace{\frac{\partial^2 \ln x}{\partial \ln \sigma_{ij} \partial \ln MC_{zi}}}_{\text{Interaction term}}.$$

Note, that if $\frac{\partial^2 \ln x}{\partial \ln \sigma_{ij} \partial \ln MC_{zi}} > 0$, reductions in σ_{ij} (FDI barriers) and MC_{zi} (intangible costs) are complements: their joint reduction amplifies x superlinearly. This aligns with the following proposition stating that globalization ($\sigma_{ij} \uparrow$) and cheaper intangibles ($MC_{zi} \downarrow$) mutually reinforce intangible accumulation.

Proposition 3 (Scallable Complementarity). *Reductions in FDI barriers ($\sigma_{ij} \uparrow$) and intangible production costs ($MC_{zi} \downarrow$) are complementary: their joint reduction generates a superlinear increase in intangible intensity (s_i^z) and expansion propensity (p_i). Formally, for $\theta \in [0, 1]$, the combined elasticity with respect to σ_{ij} and MC_{zi} satisfies:*

$$\varepsilon_{x,(\sigma_{ij} \cdot MC_{zi})} > \varepsilon_{x,\sigma_{ij}} + \varepsilon_{x,MC_{zi}},$$

where $x \in \{s_i^z, p_i\}$. Moreover, the degree of complementarity rises with θ .

This complementarity reflects the mutual reinforcement between globalization, represented by reductions in FDI barriers (σ_{ij}), and technological advancements, captured by lower intangible production costs (MC_{zi}). Nonrival intangibles ($\theta > 0$) are particularly scalable, allowing firms to deploy the same intangible asset, such as software or patents, across multiple markets at minimal marginal cost. When MC_{zi} decreases, firms are incentivized to invest more in intangible capital, as these assets yield long-term scalable returns. Simultaneously, a reduction in σ_{ij} enhances market access, increasing the marginal returns from these intangibles by enabling their utilization across a larger number of countries. The interaction between these forces magnifies the overall effect: as globalization expands market opportunities, the value of lower intangible costs is amplified, and vice versa. For instance, a technology firm experiencing declining software development costs and fewer regulatory hurdles can scale its products globally at near-zero additional cost, creating a feedback loop where each reduction strengthens the benefits of the other. This scalability is what drives the superlinear increase in intangible intensity and multinational expansion.

Proposition 3's predictions are borne out in the empirical trends documented in Section 2. The rise in intangible intensity (Figure 1a) and multinational production (Figure 1b) coincides with simultaneous declines in FDI barriers (e.g., trade liberalization) and intangible costs (e.g., digitalization). The cross-sectional evidence (Table 1) reveals that MNEs operating in more countries exhibit disproportionately higher intangible intensity, consistent with the complementarity mechanism: firms that expand globally ($\sigma_{ij} \downarrow$) invest more in scalable intangibles ($MC_{zi} \downarrow$). The event study (Figure 3) further corroborates this: firms becoming MNEs experience persistent increases in intangible intensity, reflecting the synergy between foreign expansion and intangible accumulation. Additionally, the within-industry growth in intangible capital and the rightward shift in its distribution (Figure 2a) cannot be explained by sectoral composition, pointing to firm-level responses to falling σ_{ij} and MC_{zi} . Finally, the post-2005 acceleration in MNE intangible intensity (Table 1, interaction terms) aligns with the proliferation of digital technologies ($MC_{zi} \downarrow$) and globalization ($\sigma_{ij} \downarrow$), underscoring the superlinear gains predicted by Proposition 3. Together, the evidence highlights how nonrivalry transforms cost reductions and market access into mutually reinforcing drivers of intangible-intensive globalization.

Proposition 4 (Tax Incentives, Enforcement, and Nonrivalry). *Let τ_i denote the corporate tax rate in region i , τ_H the tax-haven rate ($\tau_H < \tau_i$), and $C_i(\lambda)$ the cost of shifting profits. Then:*

1. *The elasticities of intangible intensity (s_i^z), expansion propensity (p_i), and profit shifting (ps_i)*

with respect to tax differentials $(\tau_i - \tau_H)$ increase with the degree of nonrivalry θ .

2. Strengthening tax enforcement (raising $C_i(\lambda)$) reduces s_i^z , p_i , and ps_i , with effects increasing with the degree of nonrivalry θ .

Formally:

$$\frac{\partial |\varepsilon_{x,(\tau_i - \tau_H)}|}{\partial \theta} > 0, \quad \frac{\partial x}{\partial C_i(\lambda)} < 0, \quad \frac{\partial^2 x}{\partial C_i(\lambda) \partial \theta} < 0.$$

where $x \in \{s_i^z, p_i, ps_i\}$.

Proposition 4 captures the interplay between tax policy, enforcement, and the nonrival nature of intangible capital. Firms with highly nonrival intangibles ($\theta \rightarrow 1$) face stronger incentives to shift profits to low-tax jurisdictions because their scalable assets (e.g., software, patents) can generate income across multiple locations with minimal marginal cost. A larger tax differential $(\tau_i - \tau_H)$ magnifies these incentives: the returns to shifting a unit of intangible capital grow with θ , as the same asset can be leveraged globally. Conversely, heightened tax enforcement (raising $C_i(\lambda)$) disrupts this mechanism by increasing the cost of profit shifting. Critically, the impact of both tax differentials and enforcement is asymmetric: high- θ firms are more responsive because their profitability hinges on scalable intangibles, which require profit shifting to offset upfront R&D costs. For example, a tech firm ($\theta \approx 1$) facing weaker tax incentives post-TCJA would sharply reduce intangible investment and expansion, while a manufacturer ($\theta \approx 0$) would respond modestly. Proposition 4 thus highlights how nonrivalry amplifies the tax sensitivity of intangible-intensive firms.

Proposition 4's predictions are strongly supported by the empirical evidence in Section 2. The rise in profit shifting firms (Figure 1d) from 55% to 74% (1995–2017) and its subsequent decline post-TCJA align with the model's emphasis on tax differentials $(\tau_i - \tau_H)$ as a driver of intangible intensity (s_i^z) and profit shifting (ps_i). Cross-sectional regressions (Table 1) reveal that profit-shifting MNEs exhibit 8.1 percentage points higher intangible intensity than non-shifting firms, consistent with the mechanism where high- θ firms exploit tax differentials to scale intangibles. The event study (Figure 3) further validates this: firms initiating profit-shifting activities experience rising intangible intensity, reflecting the predicted feedback loop between tax optimization and intangible accumulation. The disproportionate post-2005 rise in intangible intensity for profit-shifting MNEs (Table 1, interaction terms) coincides with digitalization-driven declines in $C_i(\lambda)$ (e.g., easier cross-border IP transfers), underscoring how enforcement costs shape behavior in high- θ sectors.

Proposition 5. *Reductions in FDI barriers ($\sigma_{ij} \downarrow$) and weaker tax enforcement ($\mathcal{C}_i(\lambda) \downarrow$) generate superlinear increases in: (i) Intangible intensity (s_i^z), (ii) Expansion propensity (p_i), (iii) Profit shifting (ps_i), with complementarity rising in nonrivalry (θ). Formally, for $x \in \{s_i^z, p_i, ps_i\}$:*

$$\varepsilon_{x,(\sigma_{ij},\mathcal{C}_i(\lambda))} = \underbrace{\frac{\partial \ln x}{\partial \ln \sigma_{ij}}}_{\text{Elasticity to FDI barriers}} + \underbrace{\frac{\partial \ln x}{\partial \ln \mathcal{C}_i(\lambda)}}_{\text{Elasticity to enforcement}} + \underbrace{\frac{\partial^2 \ln x}{\partial \ln \sigma_{ij} \partial \ln \mathcal{C}_i(\lambda)}}_{\text{Interaction term}} > \varepsilon_{x,\sigma_{ij}} + \varepsilon_{x,\mathcal{C}_i(\lambda)}.$$

and $\frac{\partial}{\partial \theta} \left(\frac{\partial^2 \ln x}{\partial \ln \sigma_{ij} \partial \ln \mathcal{C}_i(\lambda)} \right) > 0$.

The complementarity identified in Proposition 5 stems from the interplay between globalization and tax policy, operating through three interconnected mechanisms: intangible intensity, multinational expansion, and profit shifting. Reductions in FDI barriers ($\sigma_{ij} \downarrow$) lower the costs of establishing foreign subsidiaries, enabling firms to expand their global footprint. This geographic expansion amplifies the returns to intangible capital (s_i^z) by allowing firms to deploy nonrival assets—such as patents or software—across more locations simultaneously. At the same time, weaker tax enforcement ($\mathcal{C}_i(\lambda) \downarrow$) reduces the risks and costs associated with reallocating profits to low-tax jurisdictions. These two forces interact synergistically: broader multinational networks increase the scale over which intangibles can generate shifted profits, while lax enforcement lowers the marginal cost of shifting each unit of income.

For firms with highly nonrival intangibles ($\theta \approx 1$), this synergy is particularly potent. The ability to reuse intangible assets across subsidiaries at near-zero marginal cost magnifies both the returns to global expansion and the benefits of profit shifting. For example, a technology firm can invest in R&D to develop software ($s_i^z \uparrow$), establish subsidiaries in jurisdictions with lower trade barriers ($p_i \uparrow$), and license its intellectual property to affiliates in tax havens at below-market rates ($ps_i \uparrow$). Lower FDI barriers facilitate the proliferation of subsidiaries, while weaker enforcement ensures that profit-shifting strategies remain financially viable.

Critically, the interaction between these policies generates superlinear effects. Reductions in FDI barriers enhance the marginal return to weaker enforcement by expanding the network of subsidiaries through which profits can be shifted. Conversely, weaker enforcement amplifies the marginal return to globalization by increasing the after-tax value of intangible capital deployed across borders. This feedback loop disproportionately benefits firms with scalable intangibles ($\theta \uparrow$), as their profitability hinges on exploiting cross-border scalability. The result is a self-reinforcing cycle where falling barriers and enforcement costs jointly drive growth in intangible investment,

multinational activity, and profit shifting—with effects that compound in sectors characterized by high nontrivalry.

5 Quantitative Results

TO BE COMPLETED

6 Related Literature

TO BE COMPLETED

7 Conclusion

This paper demonstrates that the rise of intangible capital, the expansion of multinational enterprises (MNEs), and the increasing prevalence of international profit shifting are deeply interconnected trends. Using firm-level data, we document strong empirical correlations between intangible intensity, multinational activity, and profit shifting, motivating the need for a structural framework to disentangle the mechanisms underlying these relationships. We develop a dynamic general equilibrium model in which firms endogenously choose to invest in intangible capital, expand internationally, and reallocate profits across jurisdictions. The model identifies scalable complementarity as a key driver of these trends: reductions in the cost of intangible capital and trade barriers reinforce each other, amplifying intangible investment and multinational expansion. Additionally, we show that profit shifting emerges endogenously due to the interaction of tax differentials, enforcement costs, and the nonrival nature of intangible capital. Counterfactual analyses highlight the importance of these interactions, revealing that the rise of intangible-intensive MNEs engaged in profit shifting is not driven by any single factor but rather by the interplay of technological change, globalization, and tax policy. Our findings have important implications for corporate taxation, as policies aimed at curbing profit shifting may also influence firm-level investment and multinational activity.

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Appendix

A Proofs

Proof of Proposition 1. From Lemma 2, under $\theta = 1$ (fully nonrival intangibles), equilibrium intangible capital is:

$$z \propto \left(\frac{\sum_j \mathcal{P}_{ij} \cdot \mathcal{D}_j}{MC_{zi}} \right)^\gamma, \quad \gamma = \frac{1}{1 + (\rho - 1)\alpha}, \quad (43)$$

where $\mathcal{P}_{ij} = ((1 - \sigma_{ij})A_j)^{\rho-1}$ and $\mathcal{D}_j = (Y_j^{1/\rho} P_j)$.

By Definition 2, intangible intensity is proportional to z :

$$s_i^z = \frac{z}{\sum_{j \in J_F \cup \{i\}} l_{ij}}.$$

Taking the derivative with respect to MC_{zi} :

$$\frac{\partial s_i^z}{\partial MC_{zi}} \propto \frac{\partial z}{\partial MC_{zi}} < 0 \quad (\text{since } z \propto MC_{zi}^{-\gamma}).$$

The elasticity is:

$$\varepsilon_{s_i^z, MC_{zi}} = \frac{\partial \ln s_i^z}{\partial \ln MC_{zi}} = -\gamma < 0. \quad (44)$$

From Equation (39), expansion propensity depends on profit differentials:

$$p_i = \frac{\exp\left(\frac{\pi_i^G(a, 1) - \pi_i^G(a, \emptyset)}{\sigma_\kappa}\right)}{1 + \exp\left(\frac{\pi_i^G(a, 1) - \pi_i^G(a, \emptyset)}{\sigma_\kappa}\right)}.$$

Profit differentials $\pi_i^G(a, 1) - \pi_i^G(a, \emptyset)$ depend on z (via Equations 16–19). Since $z \downarrow$ as $MC_{zi} \uparrow$, we have:

$$\frac{\partial p_i}{\partial MC_{zi}} < 0.$$

For $\theta > 0$, the term $\sum_j \mathcal{P}_{ij} \cdot \mathcal{D}_j$ in (43) grows due to nonrival deployment (scalability). This amplifies the sensitivity of z to MC_{zi} :

$$\frac{\partial |\varepsilon_{s_i^z, MC_{zi}}|}{\partial \theta} > 0, \quad \frac{\partial |\varepsilon_{p_i, MC_{zi}}|}{\partial \theta} > 0.$$

□

Proof of Proposition 2. From Lemma 2, equilibrium intangible capital under $\theta = 1$ is:

$$z \propto \left(\frac{\sum_j \mathcal{P}_{ij} \cdot \mathcal{D}_j}{MC_{zi}} \right)^\gamma, \quad \gamma = \frac{1}{1 + (\rho - 1)\alpha}, \quad (45)$$

where $\mathcal{P}_{ij} = ((1 - \sigma_{ij})A_j)^{\rho-1}$. Higher FDI barriers ($\sigma_{ij} \uparrow$) reduce $\mathcal{P}_{ij}$, lowering z .

By Definition 2, $s_i^z \propto z$. Taking derivatives with respect to σ_{ij} and κ_{ij} :

$$\frac{\partial s_i^z}{\partial \sigma_{ij}} \propto \frac{\partial z}{\partial \sigma_{ij}} < 0, \quad \frac{\partial s_i^z}{\partial \kappa_{ij}} \propto \frac{\partial z}{\partial \kappa_{ij}} < 0.$$

Elasticities are:

$$\varepsilon_{s_i^z, \sigma_{ij}} = \frac{\partial \ln s_i^z}{\partial \ln \sigma_{ij}} = \frac{(\rho - 1)\sigma_{ij}}{1 - \sigma_{ij}} > 0 \quad (\text{but } \sigma_{ij} \uparrow \text{ reduces } z),$$

$$\varepsilon_{s_i^z, \kappa_{ij}} = \frac{\partial \ln s_i^z}{\partial \ln \kappa_{ij}} < 0.$$

From Equation (39), p_i depends on profit differentials:

$$\pi_i^G(a, 1) - \pi_i^G(a, \emptyset) \propto \underbrace{\left(\frac{(1 - \sigma_{ij})A_j}{MC_{zi}} \right)^{\rho-1}}_{\text{Term affected by } \sigma_{ij}} - \kappa_{ij}.$$

Higher σ_{ij} or κ_{ij} reduces profit differentials, lowering p_i :

$$\frac{\partial p_i}{\partial \sigma_{ij}} < 0, \quad \frac{\partial p_i}{\partial \kappa_{ij}} < 0.$$

Higher θ amplifies the term $\sum_j \mathcal{P}_{ij} \cdot \mathcal{D}_j$ in (45), making z more sensitive to σ_{ij} and κ_{ij} :

$$\frac{\partial |\varepsilon_{s_i^z, \sigma_{ij}}|}{\partial \theta} > 0, \quad \frac{\partial |\varepsilon_{s_i^z, \kappa_{ij}}|}{\partial \theta} > 0.$$

□

Proof of Proposition 3. Under $\theta = 1$ (fully nontrivial intangibles), the equilibrium intangible capital z from Lemma 2 simplifies to:

$$z \propto \left(\frac{\sum_j \mathcal{P}_{ij} \cdot \mathcal{D}_j}{MC_{zi}} \right)^{\frac{1}{1+(\rho-1)\alpha}}, \quad (46)$$

where $\mathcal{P}_{ij} = ((1 - \sigma_{ij})A_j)^{(\rho-1)}$. Intangible intensity s_i^z is proportional to z (Definition 2). Individual elasticities are:

$$\varepsilon_{s_i^z, \sigma_{ij}} = \frac{\partial \ln s_i^z}{\partial \ln \sigma_{ij}} = \frac{(\rho - 1)\sigma_{ij}}{1 - \sigma_{ij}} > 0, \quad (47)$$

$$\varepsilon_{s_i^z, MC_{zi}} = \frac{\partial \ln s_i^z}{\partial \ln MC_{zi}} = -\frac{1}{1 + (\rho - 1)\alpha} < 0. \quad (48)$$

The interaction term is:

$$\begin{aligned} \frac{\partial^2 \ln s_i^z}{\partial \ln \sigma_{ij} \partial \ln MC_{zi}} &= \frac{\partial}{\partial \ln MC_{zi}} \left(\frac{(\rho - 1)\sigma_{ij}}{1 - \sigma_{ij}} \right) \\ &= \frac{(\rho - 1)\sigma_{ij}}{(1 - \sigma_{ij})^2} > 0. \end{aligned} \quad (49)$$

This positive cross-term confirms complementarity. The combined elasticity of s_i^z with respect to σ_{ij} and MC_{zi} is:

$$\varepsilon_{s_i^z, (\sigma_{ij} \cdot MC_{zi})} = \varepsilon_{s_i^z, \sigma_{ij}} + \varepsilon_{s_i^z, MC_{zi}} + \frac{\partial^2 \ln s_i^z}{\partial \ln \sigma_{ij} \partial \ln MC_{zi}}.$$

Substituting from (47), (48), and (49):

$$\varepsilon_{s_i^z, (\sigma_{ij} \cdot MC_{zi})} > \varepsilon_{s_i^z, \sigma_{ij}} + \varepsilon_{s_i^z, MC_{zi}}.$$

This inequality establishes superlinearity. Analogous steps hold for p_i . □

Proof of Proposition 4. From Lemma 2, equilibrium intangible capital z depends on:

$$\Omega_{ij}(\lambda) = 1 + \alpha \left(\frac{(\tau_j - \tau_i) + (\tau_i - \tau_H)(1 - \mu)\lambda - (1 - \tau_i)W_i \mathcal{C}(\lambda)}{1 - \tau_j} \right)$$

where $(\tau_i - \tau_H)$ directly scales profit-shifting incentives. Elasticity derivation:

$$\varepsilon_{s_i^z, (\tau_i - \tau_H)} = \frac{\partial \ln z}{\partial \ln(\tau_i - \tau_H)} \propto \frac{(\tau_i - \tau_H)}{1 + (\rho - 1)\alpha} \cdot \theta$$

The term θ enters multiplicatively due to the aggregator constraint:

$$z \geq \left(\sum_j z_{ij}^{\frac{1}{1-\theta}} \right)^{1-\theta}$$

The intuition here is that higher θ enables reuse of z across subsidiaries, amplifying marginal returns to tax arbitrage. Firms with scalable intangibles ($\theta \uparrow$) therefore exhibit greater sensitivity to tax differentials.

Comparative statics for enforcement costs:

$$\frac{\partial s_i^z}{\partial \mathcal{C}_i(\lambda)} \propto -\theta \cdot \frac{(1 - \tau_i)W_i}{(1 - \tau_j)}$$

The negative relationship strengthens with θ because scalable intangibles require profit shifting to offset upfront R&D costs. Cross-partial derivative shows asymmetric impact:

$$\frac{\partial^2 s_i^z}{\partial \mathcal{C}_i(\lambda) \partial \theta} < 0$$

Higher θ makes firms more vulnerable to enforcement shocks. □

Proof of Proposition 5. For $x \in \{s_i^z, p_i, ps_i\}$, express $\ln x$ in terms of $\ln z$:

$$\ln x = \eta_x \ln z + \text{constants}, \quad \eta_x = \begin{cases} 1 & (x = s_i^z) \\ \frac{\rho-1}{\rho} & (x = p_i) \\ 2 & (x = ps_i) \end{cases}.$$

The combined elasticity satisfies:

$$\varepsilon_{x, (\sigma_{ij}, \mathcal{C}_i(\lambda))} = \eta_x \left(\varepsilon_{z, \sigma_{ij}} + \varepsilon_{z, \mathcal{C}_i(\lambda)} + \frac{\partial^2 \ln z}{\partial \ln \sigma_{ij} \partial \ln \mathcal{C}_i(\lambda)} \right).$$

From Lemma 2:

$$\varepsilon_{z, \sigma_{ij}} = -\frac{(\rho - 1)\sigma_{ij}}{1 - \sigma_{ij}}, \quad \varepsilon_{z, \mathcal{C}_i(\lambda)} = -\frac{\alpha(1 - \tau_i)W_i \mathcal{C}_i(\lambda)}{1 - \tau_j}.$$

Then cross-partial derivative of z

$$\frac{\partial^2 \ln z}{\partial \ln \sigma_{ij} \partial \ln \mathcal{C}_i(\lambda)} = \frac{(\rho - 1)\alpha\theta}{(1 - \sigma_{ij})(1 - \tau_j)} > 0.$$

Substitute into the combined elasticity:

$$\varepsilon_{x,(\sigma_{ij}, \mathcal{C}_i(\lambda))} = \eta_x \left(-\frac{(\rho - 1)\sigma_{ij}}{1 - \sigma_{ij}} - \frac{\alpha(1 - \tau_i)W_i \mathcal{C}_i(\lambda)}{1 - \tau_j} + \frac{(\rho - 1)\alpha\theta}{(1 - \sigma_{ij})(1 - \tau_j)} \right).$$

Since $\eta_x > 0$ and $\theta > 0$:

$$\varepsilon_{x,(\sigma_{ij}, \mathcal{C}_i(\lambda))} > \varepsilon_{x, \sigma_{ij}} + \varepsilon_{x, \mathcal{C}_i(\lambda)}.$$

The interaction term scales with θ :

$$\frac{\partial}{\partial \theta} \left(\frac{\partial^2 \ln x}{\partial \ln \sigma_{ij} \partial \ln \mathcal{C}_i(\lambda)} \right) = \eta_x \cdot \frac{(\rho - 1)\alpha}{(1 - \sigma_{ij})(1 - \tau_j)} > 0.$$

□